



BITS Pilani

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AIML for Robotics

Module 3

Localization

Section Faculty:

Raja vadhana Prabhakar

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☐ Introduction

- ☐ Types of Robots
- ☐ Mobile Robots – Use case of Programmable solutions

☐ Components of Programmable Robotic System

- ☐ Perception
- ☐ Representation
- ☐ Motion

☐ Localization

- ☐ Taxonomy
- ☐ Types

☐ Mapping

- ☐ Feature Based
- ☐ Grid Based

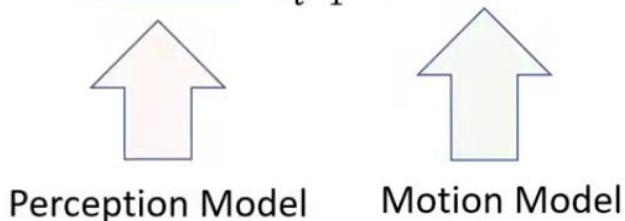
☐ Path Planning

- ☐ Graph Based
- ☐ Sample Based
- ☐ Evolutional Techniques

☐ Human – Robot Interaction

- ☐ Ethics
- ☐ Introduction to Visual Bot
- ☐ Introduction to Chat Bot

Particle Filter Approach

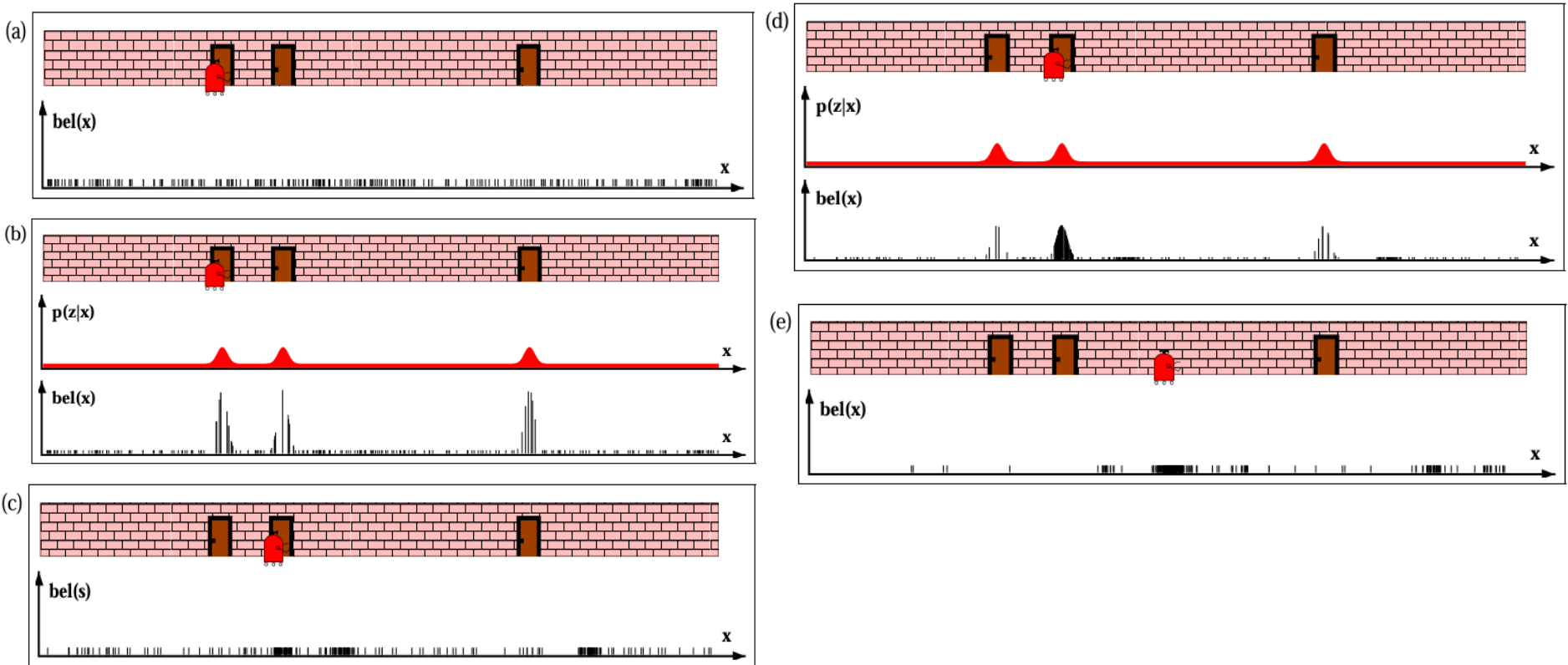
$$Bel(x_t) = \eta \boxed{P(z_t|x_t)} \int_{x_{t-1}} \boxed{P(x_t|u_t, x_{t-1})} Bel(x_{t-1}) dx_{t-1}$$


Perception Model Motion Model

Monte Carlo Based Localization



Particle Filter Approach



Particle Filter Approach

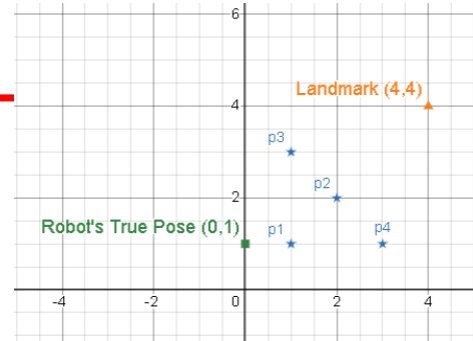
- | | |
|---------------------------------|--|
| 1. Initialize | 1. Initialize M Particles P_i with uniform belief |
| 2. Predict : Motion Update | 2. Move each P_i based on robot motion model |
| 3. Correct : Measurement Update | 3. Compare predicted sensor reading Z_{expected} with actual sensor reading Z_{measured} |
| 4. Preprocess | 4. Optionally filter out surprising measurements |
| 5. Weigh | 5. Weigh each P_i based on importance weights |
| 6. Resample | 6. Resample for particle set for next iteration based on Darwinian approach |
| 7. Repeat | 7. Reset the belief of particles with equal likelihood |

Monte Carlo Based Localization



MCL Algorithm

1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, Z_m = 5m
2. Predict : Motion Update
3. Correct : Measurement Update
4. Preprocess
5. Weigh
6. Resample
7. Repeat



	P1	P2	P3	P4
Pose X_0	(1,1,0)	(2,2,0)	(3,1,0)	(1,3,0)
Bel(X_0^m)	0.25	0.25	0.25	0.25

```

1: Algorithm MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
2:    $\hat{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$ 
3:   for  $m = 1$  to  $M$  do
4:      $x_t^{[m]} = \text{sample\_motion\_model}(u_t, x_{t-1}^{[m]})$ 
5:      $w_t^{[m]} = \text{measurement\_model}(z_t, x_t^{[m]}, m)$ 
6:      $\hat{\mathcal{X}}_t = \hat{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
7:   endfor
8:   for  $m = 1$  to  $M$  do
9:     draw  $i$  with probability  $\propto w_t^{[i]}$ 
10:    add  $x_t^{[i]}$  to  $\hat{\mathcal{X}}_t$ 
11:   endfor
12:   return  $\hat{\mathcal{X}}_t$ 

```

$$\begin{pmatrix} x' \\ y' \\ \theta' \end{pmatrix} = \begin{pmatrix} x \\ y \\ \theta \end{pmatrix} + \begin{pmatrix} -\frac{\hat{v}}{\hat{\omega}} \sin \theta + \frac{\hat{v}}{\hat{\omega}} \sin(\theta + \hat{\omega} \Delta t) \\ \frac{\hat{v}}{\hat{\omega}} \cos \theta - \frac{\hat{v}}{\hat{\omega}} \cos(\theta + \hat{\omega} \Delta t) \\ \hat{\omega} \Delta t \end{pmatrix} \quad (5.13)$$

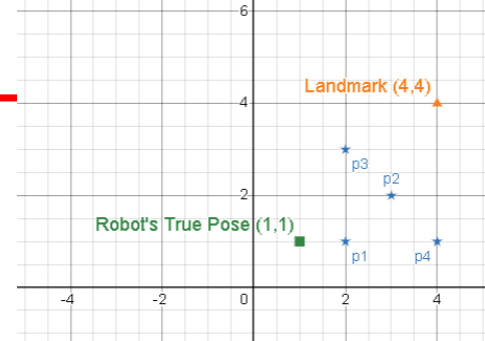
Table 8.2 MCL, or Monte Carlo Localization, a localization algorithm based on particle filters.

Monte Carlo Based Localization



MCL Algorithm

1. Initialize : $M=4$, 2D, 5×5 , $L(4.4)$, Move 1m east, Sensor noise = 0.5, $Z_m = 5m$
2. **Predict : Motion Update**
3. **Correct : Measurement Update**
4. ~~Preprocess~~
5. Weigh
6. Resample
7. Repeat



	P1	P2	P3	P4
Pose X_0	(1,1,0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(2,1,0)	(3,2,0)	(4,1,0)	(2,3,0)

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1: Algorithm MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
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6:      $\hat{\mathcal{X}}_t = \hat{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
7:   endfor
8:   for  $m = 1$  to  $M$  do
9:     draw  $i$  with probability  $\propto w_t^{[i]}$ 
10:    add  $x_t^{[i]}$  to  $\mathcal{X}_t$ 
11:   endfor
12:   return  $\mathcal{X}_t$ 
    
```

Inefficient form (modified for simplicity)

$$\begin{pmatrix} x_t \\ y_t \\ \theta_t \end{pmatrix} = \begin{pmatrix} x_{t-1} \\ y_{t-1} \\ \theta_{t-1} \end{pmatrix} + \begin{pmatrix} v \cos \theta \Delta t \\ v \sin \theta \Delta t \\ w \Delta t \end{pmatrix}$$

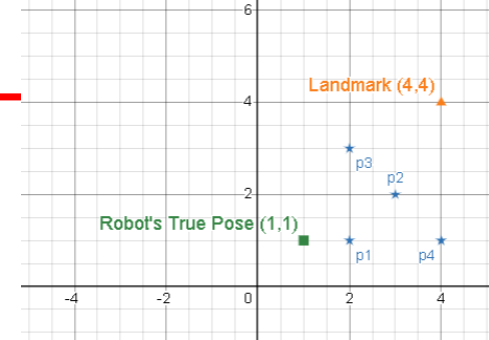
Linear Regression Model $x_t = x_{t-1} + 1$ 8

Table 8.2 MCL, or Monte Carlo Localization, a localization algorithm based on particle filters.

Monte Carlo Based Localization

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	P1	P2	P3	P4
Pose X_0	(1,1,0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(2,1,0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	3.6	2.24	3	2.24

Sample Calculation : $W_1^m = e^{-\frac{(5-3.6)^2}{2*0.5}} = \sim 0.14$. For simplicity σ^2 is assumed to be noise = 0.5. This is updated in all the upcoming slides with weight correction.

```

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```

(modified for simplicity)

$$\mathcal{N}(z_t^k; z_t^{k*}, \sigma_{\text{hit}}^2) = \frac{1}{\sqrt{2\pi\sigma_{\text{hit}}^2}} e^{-\frac{1}{2} \frac{(z_t^k - z_t^{k*})^2}{\sigma_{\text{hit}}^2}}$$

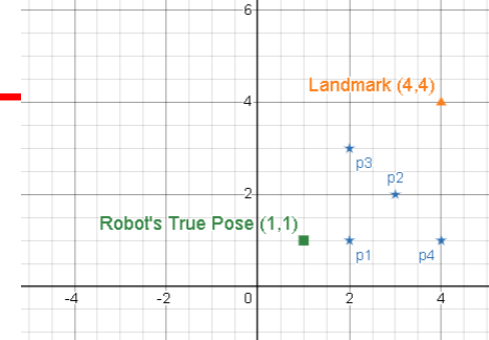
$$h_\theta(x) = \frac{1}{1 + e^{(-\theta^T x)}} \quad (b.5)$$

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Monte Carlo Based Localization

MCL Algorithm

1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, $Z_m = 5m$
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$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(2,1,0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	3.6	2.24	3	2.24
W_1^m	0.14	0.0005	0.018	0.0005

```

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```

(modified for simplicity)

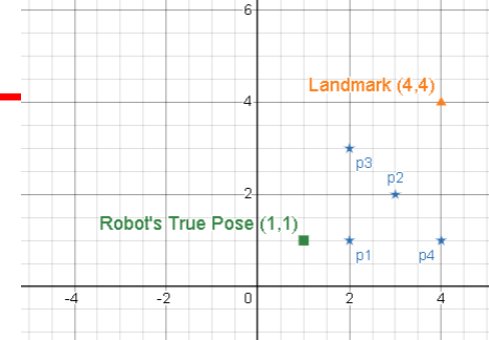
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Table 8.2 MCL, or Monte Carlo Localization, a localization algorithm based on particle filters.

Monte Carlo Based Localization

MCL Algorithm

1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, $Z_m = 5m$
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6. Resample
7. Repeat



	P1	P2	P3	P4
Pose X_0	(1,1,0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(2,1,0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	3.6	2.24	3	2.24
W_1^m	0.14	0.0005	0.018	0.0005
$\text{Bel}(X_1^m)$	0.8805	0.0013	0.1132	0.0031

(modified for simplicity)

$$\mathcal{N}(z_t^k; z_t^{k*}, \sigma_{\text{hit}}^2) = \frac{1}{\sqrt{2\pi\sigma_{\text{hit}}^2}} e^{-\frac{1}{2} \frac{(z_t^k - z_t^{k*})^2}{\sigma_{\text{hit}}^2}}$$

$$h_\theta(x) = \frac{1}{1 + e^{(-\theta^T x)}} \quad (b.5)$$

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1:  Algorithm MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
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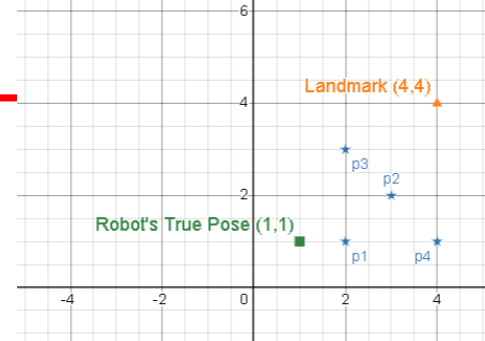
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Monte Carlo Based Localization



MCL Algorithm

1. Initialize : $M=4$, 2D, 5×5 , $L(4.4)$, Move 1m east, Sensor noise = 0.5, $Z_m = 5m$
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Pose X_0	(1,1,0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(2,1,0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	3.6	2.24	3	2.24
W_1^m	0.14	0.0005	0.018	0.0005
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```

1:  Algorithm MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
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5:       $w_t^{[m]} = \text{measurement\_model}(z_t, x_t^{[m]}, m)$ 
6:       $\hat{\mathcal{X}}_t = \hat{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
7:    endfor
8:    for  $m = 1$  to  $M$  do
9:      draw  $i$  with probability  $\propto w_t^{[i]}$ 
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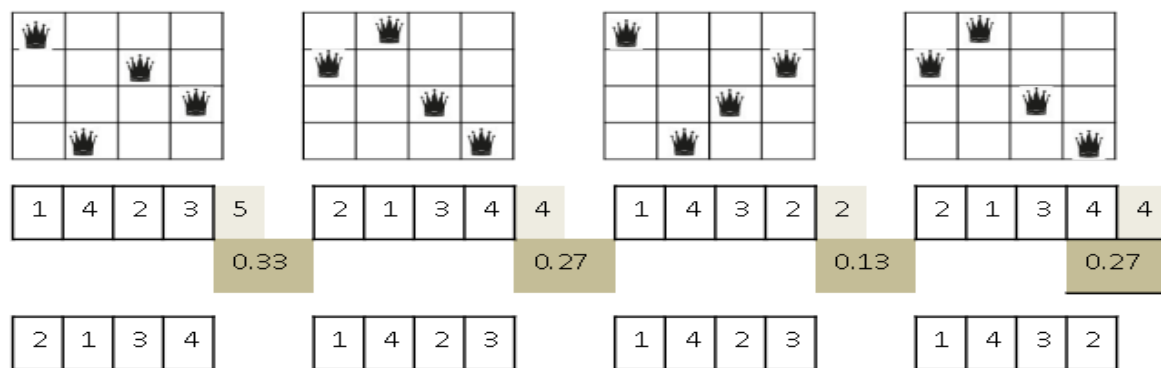
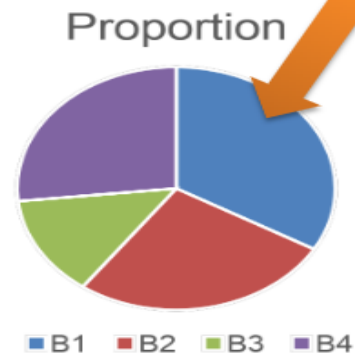
```

Table 8.2 MCL, or Monte Carlo Localization, a localization algorithm based on particle filters.

Genetic Algorithm



Eg., use roulette wheel mechanism to select pair/s



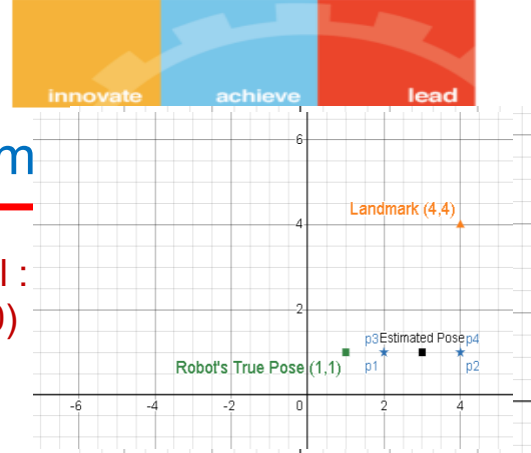
Sample winners of game -1 ,2,3,4 : B4, B1, B1, B3

Monte Carlo Based Localization

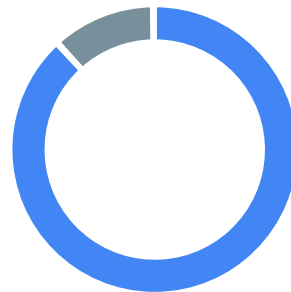
MCL Algorithm

1. Initialize : $M=4$, 2D, 5×5 , $L(4.4)$, Move 1m east, Sensor noise = 0.5, $Z_m = 5m$
2. Predict : Motion Update
3. Correct : Measurement Update
4. ~~Preprocess~~
5. Weigh
6. **Resample**
7. Repeat

One Naïve Estimation Model :
Average of Particles = $(3,1,0)$



Roulette wheel
(Similar to GA's parent selection)



■ P1 ■ P2 ■ P3 ■ P4

```

1: Algorithm MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
2:    $\mathcal{X}_t = \mathcal{X}_{t-1} = \emptyset$ 
3:   for  $m = 1$  to  $M$  do
4:      $x_t^{[m]} = \text{sample\_motion\_model}(u_t, x_{t-1}^{[m]})$ 
5:      $w_t^{[m]} = \text{measurement\_model}(z_t, x_t^{[m]}, m)$ 
6:      $\mathcal{X}_t = \mathcal{X}_t + \{x_t^{[m]}, w_t^{[m]}\}$ 
7:   endfor
8:   for  $m = 1$  to  $M$  do
9:     draw  $i$  with probability  $\propto w_t^{[i]}$ 
10:    add  $x_t^{[i]}$  to  $\mathcal{X}_t$ 
11:   endfor
12:   return  $\mathcal{X}_t$ 
    
```

	P1	P2	P3	P4
Pose X_0	(1,1,0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(2,1,0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	3.6	2.24	3	2.24
W_1^m	0.14	0.0005	0.018	0.0005
$\text{Bel}(X_1^m)$	0.8805	0.0013	0.1132	0.0031

	P1	P2	P3	P4
X_1	(2,1,0)	(4,1,0)	(4,1,0)	(2,1,0)
$\text{Bel}(X_1^m)$	0.25	0.25	0.25	0.25

Table 8.2 MCL, or Monte Carlo Localization, a localization algorithm based on particle filters.

Particle Filter Approach

- | | |
|---------------------------------|--|
| 1. Initialize | 1. Initialize M Particles P_i with uniform belief |
| 2. Predict : Motion Update | 2. Move each P_i based on robot motion model |
| 3. Correct : Measurement Update | 3. Compare predicted sensor reading Z_{expected} with actual sensor reading Z_{measured} |
| 4. Preprocess | 4. Optionally filter out surprising measurements |
| 5. Weigh | 5. Weigh each P_i based on importance weights |
| 6. Resample | 6. Resample for particle set for next iteration based on Darwinian approach |
| 7. Repeat | 7. Reset the belief of particles with equal likelihood |

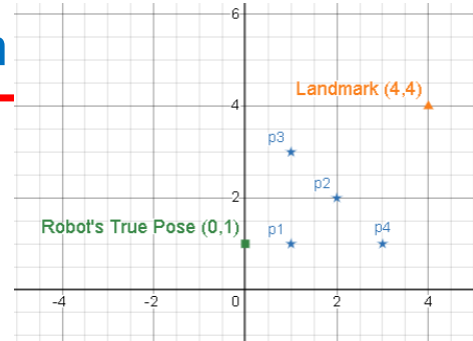
Monte Carlo Based Localization

Augmented MCL Algorithm



1. Initialize : $M=4$, 2D, 5×5 , $L(4.4)$, Move 1m east,
Sensor noise = 0.5, $Z_m = 5m$, $\alpha_{fast} = 0.5$, $\alpha_{slow} = 0.1$
2. Predict : Motion Update
3. Correct : Measurement Update
4. Preprocess
5. Weigh
6. Resample
7. Repeat

No Change



	P1	P2	P3	P4
Pose X_0	(1,1,0)	(2,2,0)	(3,1,0)	(1,3,0)
$Bel(X_0^m)$	0.25	0.25	0.25	0.25

```

1: Algorithm Augmented_MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
2:   static  $w_{slow}, w_{fast}$ 
3:    $\mathcal{X}_t = \mathcal{X}_{t-1}$ 
4:   for  $m = 1$  to  $M$  do
5:      $x_t^{[m]} = \text{sample-motion-model}(u_t, x_{t-1}^{[m]})$ 
6:      $w_t^{[m]} = \text{measurement-model}(z_t, x_t^{[m]}, m)$ 
7:      $\tilde{\mathcal{X}}_t = \mathcal{X}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
8:      $w_{avg} = w_{avg} + \frac{1}{M} w_t^{[m]}$ 
9:   endfor
10:   $w_{slow} = w_{slow} + \alpha_{slow} (w_{avg} - w_{slow})$ 
11:   $w_{fast} = w_{fast} + \alpha_{fast} (w_{avg} - w_{fast})$ 
12:  for  $m = 1$  to  $M$  do
13:    with probability  $\max(0.0, 1.0 - w_{fast}/w_{slow})$  do
14:      add random pose to  $\mathcal{X}_t$ 
15:    else
16:      draw  $i \in \{1, \dots, N\}$  with probability  $\propto w_t^{[i]}$ 
17:      add  $x_t^{[i]}$  to  $\mathcal{X}_t$ 
18:    endwith
19:  endfor
20:  return  $\mathcal{X}_t$ 
    
```

Table 8.3 An adaptive variant of MCL that adds random samples. The number of random samples is determined by comparing the short-term with the long-term likelihood of sensor measurements.

Monte Carlo Based Localization

Augmented MCL Algorithm

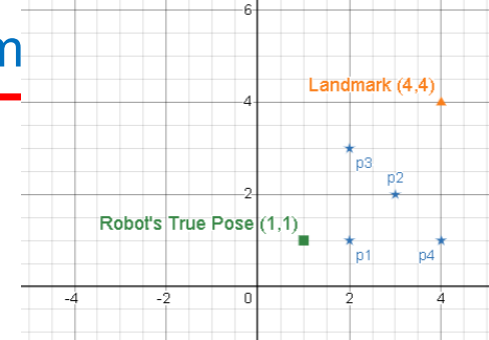
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Sensor noise = 0.5, $Z_m = 5m$, $\alpha_{fast} = 0.5$, $\alpha_{slow} = 0.1$
2. **Predict : Motion Update**
3. Correct : Measurement Update
4. ~~Preprocess~~
5. Weigh
6. Resample
7. Repeat

No Change

```

1: Algorithm Augmented_MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
2:   static  $w_{slow}, w_{fast}$ 
3:    $\tilde{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$ 
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6:      $w_t^{[m]} = \text{measurement-model}(z_t, x_t^{[m]}, m)$ 
7:      $\tilde{\mathcal{X}}_t = \tilde{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
8:      $w_{avg} = w_{avg} + \frac{1}{M} w_t^{[m]}$ 
9:   endwhile
10:   $w_{slow} = w_{slow} + \alpha_{slow} (w_{avg} - w_{slow})$ 
11:   $w_{fast} = w_{fast} + \alpha_{fast} (w_{avg} - w_{fast})$ 
12:  for  $m = 1$  to  $M$  do
13:    with probability  $\max(0.0, 1.0 - w_{fast}/w_{slow})$  do
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19:  endwhile
20:  return  $\tilde{\mathcal{X}}_t$ 
    
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	P1	P2	P3	P4
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$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(2,1,0)	(3,2,0)	(4,1,0)	(2,3,0)

(modified for simplicity)

$$\begin{pmatrix} x_t \\ y_t \\ \theta_t \end{pmatrix} = \begin{pmatrix} x_{t-1} \\ y_{t-1} \\ \theta_{t-1} \end{pmatrix} + \begin{pmatrix} v \cos \theta \Delta t \\ v \sin \theta \Delta t \\ w \Delta t \end{pmatrix}$$

Linear Regression Model $x_t = x_{t-1} + 1$ 17

Monte Carlo Based Localization

Augmented MCL Algorithm

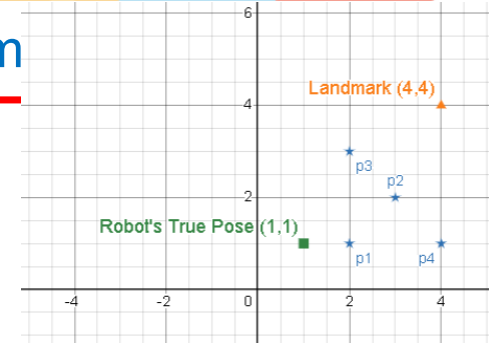
1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, $Z_m = 5m$, $\alpha_{fast} = 0.5$, $\alpha_{slow} = 0.1$
2. Predict : Motion Update
3. **Correct : Measurement Update**
4. ~~Preprocess~~
5. Weigh
6. Resample
7. Repeat

No Change

```

1: Algorithm Augmented_MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
2:   static  $w_{slow}, w_{fast}$ 
3:    $\mathcal{X}_t = \mathcal{X}_{t-1}$ 
4:   for  $m = 1$  to  $M$  do
5:      $x_t^{[m]} = \text{sample-motion-model}(u_t, x_{t-1}^{[m]})$ 
6:      $w_t^{[m]} = \text{measurement-model}(z_t, x_t^{[m]}, m)$ 
7:      $\tilde{\mathcal{X}}_t = \mathcal{X}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
8:      $w_{avg} = w_{avg} + \frac{1}{M} w_t^{[m]}$ 
9:   endfor
10:   $w_{slow} = w_{slow} + \alpha_{slow}(w_{avg} - w_{slow})$ 
11:   $w_{fast} = w_{fast} + \alpha_{fast}(w_{avg} - w_{fast})$ 
12:  for  $m = 1$  to  $M$  do
13:    with probability  $\max(0.0, 1.0 - w_{fast}/w_{slow})$  do
14:      add random pose to  $\mathcal{X}_t$ 
15:    else
16:      draw  $i \in \{1, \dots, N\}$  with probability  $\propto w_t^{[i]}$ 
17:      add  $x_t^{[i]}$  to  $\mathcal{X}_t$ 
18:    endwith
19:  endfor
20:  return  $\mathcal{X}_t$ 
    
```

Table 8.3 An adaptive variant of MCL that adds random samples. The number of random samples is determined by comparing the short-term with the long-term likelihood of sensor measurements.



	P1	P2	P3	P4
Pose X_0	(1,1,0)	(2,2,0)	(3,1,0)	(1,3,0)
Bel(X_0^m)	0.25	0.25	0.25	0.25
X_1	(2,1,0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	3.6	2.24	3	2.24

(modified for simplicity)

$$\mathcal{N}(z_t^k; z_t^{k*}, \sigma_{hit}^2) = \frac{1}{\sqrt{2\pi\sigma_{hit}^2}} e^{-\frac{1}{2} \frac{(z_t^k - z_t^{k*})^2}{\sigma_{hit}^2}}$$

$$h_\theta(x) = \frac{1}{1 + e^{(-\theta^T x)}} \quad (b.5)$$

Monte Carlo Based Localization

Augmented MCL Algorithm

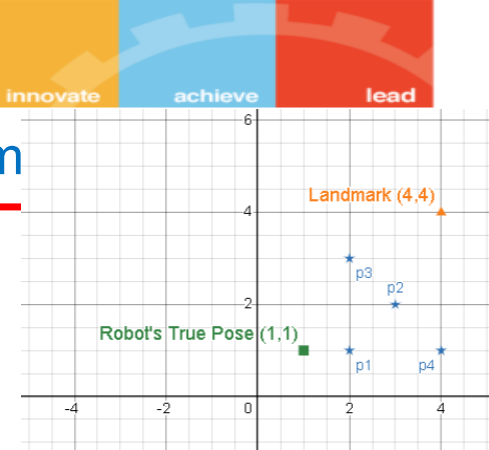
1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, $Z_m = 5m$, $\alpha_{fast} = 0.5$, $\alpha_{slow} = 0.1$
2. Predict : Motion Update
3. **Correct : Measurement Update**
4. ~~Preprocess~~
5. Weigh
6. Resample
7. Repeat

No Change

```

1: Algorithm Augmented_MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
2:   static  $w_{slow}, w_{fast}$ 
3:    $\mathcal{X}_t = \mathcal{X}_{t-1}$ 
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5:      $x_t^{[m]} = \text{sample-motion-model}(u_t, x_{t-1}^{[m]})$ 
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7:      $\tilde{\mathcal{X}}_t = \mathcal{X}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
8:      $w_{avg} = w_{avg} + \frac{1}{M} w_t^{[m]}$ 
9:   endfor
10:   $w_{slow} = w_{slow} + \alpha_{slow}(w_{avg} - w_{slow})$ 
11:   $w_{fast} = w_{fast} + \alpha_{fast}(w_{avg} - w_{fast})$ 
12:  for  $m = 1$  to  $M$  do
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16:      draw  $i \in \{1, \dots, N\}$  with probability  $\propto w_t^{[i]}$ 
17:      add  $x_t^{[i]}$  to  $\mathcal{X}_t$ 
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20:  return  $\mathcal{X}_t$ 
    
```

Table 8.3 An adaptive variant of MCL that adds random samples. The number of random samples is determined by comparing the short-term with the long-term likelihood of sensor measurements.



	P1	P2	P3	P4
Pose X_0	(1,1,0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(2,1,0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	3.6	2.24	3	2.24
W_1^m	0.14	0.0005	0.018	0.0005

(modified for simplicity)

$$\mathcal{N}(z_t^k; z_t^{k*}, \sigma_{hit}^2) = \frac{1}{\sqrt{2\pi\sigma_{hit}^2}} e^{-\frac{1}{2} \frac{(z_t^k - z_t^{k*})^2}{\sigma_{hit}^2}} \quad (6.5)$$

Monte Carlo Based Localization

Augmented MCL Algorithm

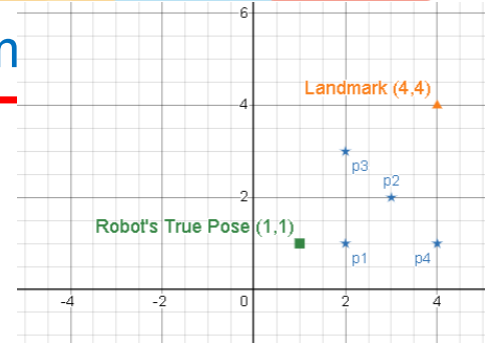
1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, $Z_m = 5m$, $\alpha_{fast} = 0.5$, $\alpha_{slow} = 0.1$
2. Predict : Motion Update
3. **Correct : Measurement Update**
4. ~~Preprocess~~
5. **Weigh**
6. Resample
7. Repeat

No Change

```

1: Algorithm Augmented_MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
2:   static  $w_{slow}, w_{fast}$ 
3:    $\mathcal{X}_t = \mathcal{X}_{t-1}$ 
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5:      $x_t^{[m]} = \text{sample-motion-model}(u_t, x_{t-1}^{[m]})$ 
6:      $w_t^{[m]} = \text{measurement-model}(z_t, x_t^{[m]}, m)$ 
7:      $\tilde{\mathcal{X}}_t = \mathcal{X}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
8:      $w_{avg} = w_{avg} + \frac{1}{M} w_t^{[m]}$ 
9:   endfor
10:   $w_{slow} = w_{slow} + \alpha_{slow} (w_{avg} - w_{slow})$ 
11:   $w_{fast} = w_{fast} + \alpha_{fast} (w_{avg} - w_{fast})$ 
12:  for  $m = 1$  to  $M$  do
13:    with probability  $\max(0.0, 1.0 - w_{fast}/w_{slow})$  do
14:      add random pose to  $\mathcal{X}_t$ 
15:    else
16:      draw  $i \in \{1, \dots, N\}$  with probability  $\propto w_t^{[i]}$ 
17:      add  $x_t^{[i]}$  to  $\mathcal{X}_t$ 
18:    endwith
19:  endfor
20:  return  $\mathcal{X}_t$ 
    
```

Table 8.3 An adaptive variant of MCL that adds random samples. The number of random samples is determined by comparing the short-term with the long-term likelihood of sensor measurements.



	P1	P2	P3	P4
Pose X_0	(1,1,0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(2,1,0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	3.6	2.24	3	2.24
W_1^m	0.14	0.0005	0.018	0.0005
$\text{Bel}(X_1^m)$	0.8805	0.0013	0.1132	0.0031

(modified for simplicity)

$$\mathcal{N}(z_t^k; z_t^{k*}, \sigma_{hit}^2) = \frac{1}{\sqrt{2\pi\sigma_{hit}^2}} e^{-\frac{1}{2} \frac{(z_t^k - z_t^{k*})^2}{\sigma_{hit}^2}}$$

$$h_\theta(x) = \frac{1}{1 + e^{(-\theta^T x)}} \quad (b.5)$$

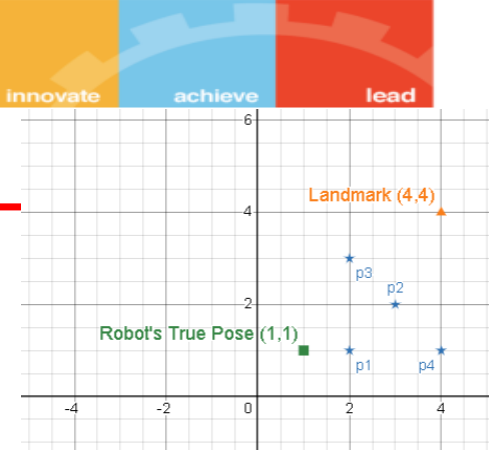
Monte Carlo Based Localization

Note : Changes are made to illustrate both the process. Assume Short term likelihood = 0.5 instead of 0 and long term likelihood of 0.9 instead of 0. Since initial values used in live class room discussion was not yielding the expected inference this is updated.

Augmented MCL Algorithm

1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, $Z_m = 5m$, $\alpha_{fast} = 0.5$, $\alpha_{slow} = 0.1$
2. Predict : Motion Update
3. Correct : Measurement Update
4. Preprocess
5. Weigh
6. Resample
7. Repeat

One Naïve Estimation Model :
Average of Particles = (3,1,0)



$$W_{avg}^M = \frac{\sum_1^M W(X_1^i)}{4} = \sim 0.03975$$

	P1	P2	P3	P4
Pose X_0	(1,1,0)	(2,2,0)	(3,1,0)	(1,3,0)
Bel(X_0^m)	0.25	0.25	0.25	0.25
X_1	(2,1,0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	3.6	2.24	3	2.24
W_1^m	0.14	0.0005	0.018	0.0005
Bel(X_1^m)	0.8805	0.0013	0.1132	0.0031

```

1: Algorithm Augmented_MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
2:   static  $w_{slow}, w_{fast}$ 
3:    $\mathcal{X}_t = \mathcal{X}_t = \emptyset$ 
4:   for  $m = 1$  to  $M$  do
5:      $x_t^{[m]} = \text{sample-motion-model}(u_t, x_{t-1}^{[m]})$ 
6:      $w_t^{[m]} = \text{measurement-model}(z_t, x_t^{[m]}, m)$ 
7:      $\tilde{\mathcal{X}}_t = \tilde{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
8:      $w_{avg} = w_{avg} + \frac{1}{M} w_t^{[m]}$ 
9:   endfor
10:   $w_{slow} = w_{slow} + \alpha_{slow} (w_{avg} - w_{slow})$ 
11:   $w_{fast} = w_{fast} + \alpha_{fast} (w_{avg} - w_{fast})$ 
12:  for  $m = 1$  to  $M$  do
13:    with probability  $\max(0.0, 1.0 - w_{fast}/w_{slow})$  do
14:      add random pose to  $\mathcal{X}_t$ 
15:    else
16:      draw  $i \in \{1, \dots, N\}$  with probability  $\propto w_t^{[i]}$ 
17:      add  $x_t^{[i]}$  to  $\mathcal{X}_t$ 
18:    endwith
19:  endfor
20:  return  $\mathcal{X}_t$ 

```

$$W_{fast} = (1 - \alpha_{fast}) W_{fast} + \alpha_{fast} W_{avg}$$

$$W_{fast} = (1 - 0.5)(0.5) + (0.5)(0.03975) = \sim 0.27$$

$$W_{slow} = (1 - \alpha_{slow}) W_{slow} + \alpha_{slow} W_{avg}$$

$$W_{slow} = (1 - 0.1)(0.9) + (0.1)(0.03975) = \sim 0.81$$

Inference : Approx. 30% of Particles are Informative. Retain them . Need randomization for other 70% of M particles!

$$\text{Likelihood Ratio} = \frac{0.27}{0.81} = 0.33$$

$$\text{MAX}\{0, (1 - 0.33)\} = \sim 0.67$$

Hence for M=4 number of particle selection in next round with 0.67 probability (approx. to 70 only for this slide's example), exploration proceeds and for rest exploitation is preferred

Table 8.3 An adaptive variant of MCL that adds random samples. The number of random samples is determined by comparing the short-term with the long-term likelihood of sensor measurements.

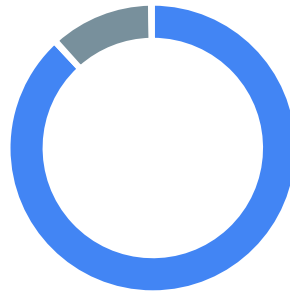
Monte Carlo Based Localization

Augmented MCL Algorithm

1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, $Z_m = 5m$, $\alpha_{fast} = 0.5$, $\alpha_{slow} = 0.1$
2. Predict : Motion Update
3. Correct : Measurement Update
4. Preprocess
5. Weigh
6. Resample
7. Repeat

One Naïve Estimation Model :
Average of Particles = (2,1,0)

Roulette wheel
(Similar to GA 's parent selection)

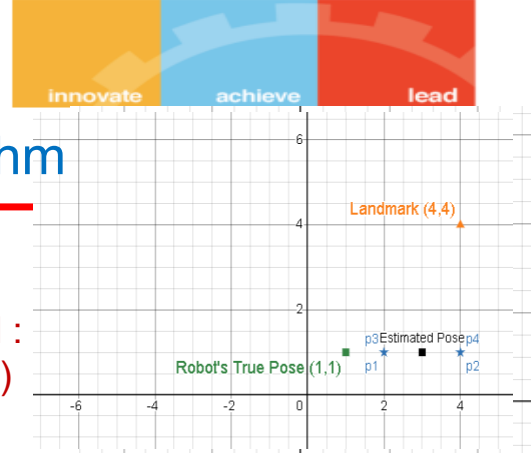


■ P1 ■ P2 ■ P3 ■ P4

```

1: Algorithm Augmented_MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
2:   static  $w_{slow}, w_{fast}$ 
3:    $\mathcal{X}_t = \mathcal{X}_{t-1}$ 
4:   for  $m = 1$  to  $M$  do
5:      $x_t^{[m]} = \text{sample-motion-model}(u_t, x_{t-1}^{[m]})$ 
6:      $w_t^{[m]} = \text{measurement-model}(z_t, x_t^{[m]}, m)$ 
7:      $\tilde{\mathcal{X}}_t = \mathcal{X}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
8:      $w_{avg} = w_{avg} + \frac{1}{M} w_t^{[m]}$ 
9:   endfor
10:   $w_{slow} = w_{slow} + \alpha_{slow} (w_{avg} - w_{slow})$ 
11:   $w_{fast} = w_{fast} + \alpha_{fast} (w_{avg} - w_{fast})$ 
12:  for  $m = 1$  to  $M$  do
13:    with probability  $\max(0.0, 1.0 - w_{fast}/w_{slow})$  do
14:      add random pose to  $\mathcal{X}_t$ 
15:    else
16:      draw  $i \in \{1, \dots, N\}$  with probability  $\propto w_t^{[i]}$ 
17:      add  $x_t^{[i]}$  to  $\mathcal{X}_t$ 
18:    endwith
19:  endfor
20:  return  $\mathcal{X}_t$ 
    
```

Based on previous assumptions note that for next iteration 3 particles are picked from random process and only one (30%) of M is from old pool ie., P1. Proceed next iteration with this resampled pool of M particles!



	P1	P2	P3	P4
Pose X_0	(1,1,0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(2,1,0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	3.6	2.24	3	2.24
W_1^m	0.14	0.0005	0.018	0.0005
$\text{Bel}(X_1^m)$	0.8805	0.0013	0.1132	0.0031

	P1	P2	P3	P4
X_1	(2,1,0)	(4,2,0)	(0,0,0)	(6,2,0)
$\text{Bel}(X_1^m)$	0.25	0.25	0.25	0.25

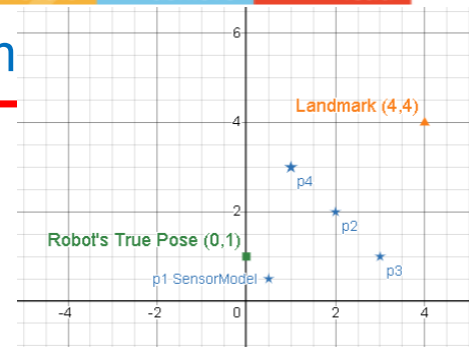
Table 8.3 An adaptive variant of MCL that adds random samples. The number of random samples is determined by comparing the short-term with the long-term likelihood of the sensor measurements.

Particle Filter Approach

1. Initialize
 2. Predict : Motion Update
 3. Correct : Measurement Update
 - ~~4. Preprocess~~
 5. Weigh
 6. Resample
 7. Repeat
1. Initialize M Particles P_i with uniform belief
 2. Move each P_i based on robot motion model
 3. Compare predicted sensor reading Z_{expected} with actual sensor reading Z_{measured}
 - ~~4. Optionally filter out surprising measurements~~
 5. Weigh each P_i based on importance weights
 6. Resample for particle set for next iteration based on Darwinian approach augmented with short term vs long term likelihood ratio
 7. Reset the belief of particles with equal likelihood

Monte Carlo Based Localization

Mixture(Dual) MCL Algorithm



1. Initialize : $M=4$, 2D, 5×5 , L(4.4), Move 1m east, Sensor noise = 0.5, $Z_m = 5m$, $\Phi = 0.25$
2. Predict
3. Correct
4. Preprocess
5. Weigh
6. Resample

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Table 3
The Mixture-MCL algorithm, here using the third variant of dual MCL (see Table 2)

Algorithm MixtureMCL(X, a, o)

```

 $X' = \emptyset$ 
generate kd-tree  $b$  from  $X$ 
for  $i = 0$  to  $m$  do
  with probability  $1 - \phi$  do
    generate random  $x$  from  $X$  according to  $w_1, \dots, w_m$ 
    generate random  $x' \sim p(x' | a, x)$ 
     $w' = p(o | x')$ 
    add  $\langle x', w' \rangle$  to  $X'$ 
  else do
    generate random  $x' \sim p(x' | o) / \pi(o)$ 
    generate random  $x \sim p(x' | \bar{a}, x) / \pi(x' | a)$ 
     $w' = \pi(o) \pi(x' | a) b(x)$ 
    add  $\langle x', w' \rangle$  to  $X'$ 
  endif
endfor
normalize the importance factors  $w'$  in  $X'$ 
return  $X'$ 
    
```

	P1	P2	P3	P4
Pose X_0	(0.5, 0.5, 0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25

Important Note :The picture with algorithm is a variation of standard Mixture MCL for reference but the given numerical problem is a flavor of vanilla Mixture MCL.

Monte Carlo Based Localization

Mixture(Dual) MCL Algorithm

1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, $Z_m = 5m$, $\Phi = 0.25$
2. **Predict**
3. **Correct**
4. **Preprocess**
5. **Weigh**
6. **Resample**

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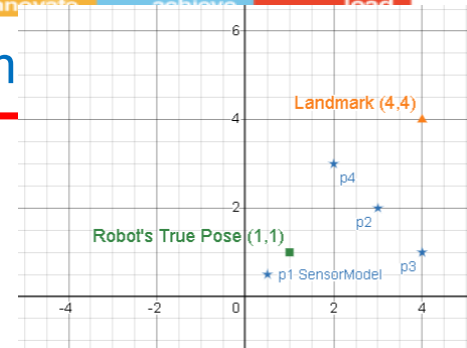
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    generate random  $x' \sim p(x' | a, x)$ 
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  else do
    generate random  $x' \sim p(x' | o) / \pi(o)$ 
    generate random  $x \sim p(x' | \bar{a}, x) / \pi(x' | a)$ 
     $w' = \pi(o) \pi(x' | a) b(x)$ 
    add  $\langle x', w' \rangle$  to  $X'$ 
  endif
endifor
normalize the importance factors  $w'$  in  $X'$ 
return  $X'$ 
    
```

P1 is drawn from the distribution of sensor measurement! Since P1 will not be affected by motion model due to swap of role, its (x,y,...) values remain same.



	P1	P2	P3	P4
Pose X_0	(0.5, 0.5, 0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(0.5, 0.5, 0)	(3,2,0)	(4,1,0)	(2,3,0)

(modified for simplicity)

$$\begin{pmatrix} x_t \\ y_t \\ \theta_t \end{pmatrix} = \begin{pmatrix} x_{t-1} \\ y_{t-1} \\ \theta_{t-1} \end{pmatrix} + \begin{pmatrix} v \cos \theta \Delta t \\ v \sin \theta \Delta t \\ w \Delta t \end{pmatrix}$$

Linear Regression Model $x_t = x_{t-1} + 1$ 25

Monte Carlo Based Localization

Mixture(Dual) MCL Algorithm

1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, $Z_m = 5m$, $\Phi = 0.25$
2. Predict
3. **Correct**
4. ~~Preprocess~~
5. Weigh
6. Resample

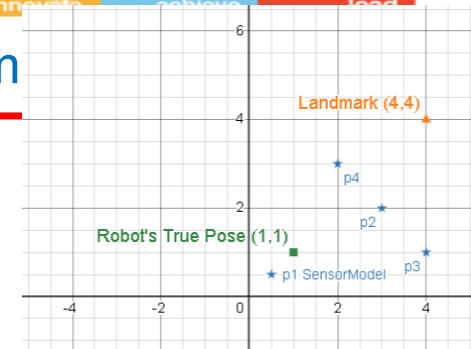
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    generate random  $x' \sim p(x' | a, x)$ 
     $w' = p(o | x')$ 
    add  $(x', w')$  to  $X'$ 
  else do
    generate random  $x' \sim p(x' | o) / \pi(o)$ 
    generate random  $x \sim p(x' | a, x) / \pi(x' | a)$ 
     $w' = \pi(o) \pi(x' | a) b(x)$ 
    add  $(x', w')$  to  $X'$ 
  endif
endfor
normalize the importance factors  $w'$  in  $X'$ 
return  $X'$ 
    
```



	P1	P2	P3	P4
Pose X_0	(0.5, 0.5, 0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(0.5, 0.5, 0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	~4.95	2.24	3	2.24

Ze is measured using this form as usual.
But for P1 alternatively you may redesign
a motion model based estimator like
this to determine $Z_e, 1$

$$\mathcal{N}(z_t^k; z_t^{k*}, \sigma_{\text{hit}}^2) = \frac{1}{\sqrt{2\pi\sigma_{\text{hit}}^2}} e^{-\frac{1}{2} \frac{(z_t^k - z_t^{k*})^2}{\sigma_{\text{hit}}^2}}$$

(modified for simplicity)

$$h_\theta(x) = \frac{1}{1 + e^{(-\theta^T x)}}$$

Monte Carlo Based Localization

Mixture(Dual) MCL Algorithm

1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, $Z_m = 5m$, $\Phi = 0.25$
2. Predict
3. **Correct**
4. ~~Preprocess~~
5. Weigh
6. Resample

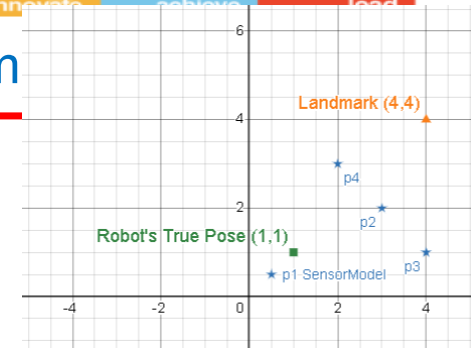
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 $X' = \emptyset$ 
generate kd-tree  $b$  from  $X$ 
for  $i = 0$  to  $m$  do
  with probability  $1 - \phi$  do
    generate random  $x$  from  $X$  according to  $w_1, \dots, w_m$ 
    generate random  $x' \sim p(x' | a, x)$ 
     $w' = p(o | x')$ 
    add  $(x', w')$  to  $X'$ 
  else do
    generate random  $x' \sim p(x' | o) / \pi(o)$ 
    generate random  $x \sim p(x' | \bar{a}, x) / \pi(x' | a)$ 
     $w' = \pi(o) \pi(x' | a) b(x)$ 
    add  $(x', w')$  to  $X'$ 
  endif
endfor
normalize the importance factors  $w'$  in  $X'$ 
return  $X'$ 
    
```



	P1	P2	P3	P4
Pose X_0	(0.5, 0.5, 0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(0.5, 0.5, 0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	~4.95	2.24	3	2.24
W_1^m	~0.995	0.0005	0.018	0.0005

(modified for simplicity)

$$\mathcal{N}(z_t^k; z_t^{k*}, \sigma_{\text{hit}}^2) = \frac{1}{\sqrt{2\pi\sigma_{\text{hit}}^2}} e^{-\frac{1}{2} \frac{(z_t^k - z_t^{k*})^2}{\sigma_{\text{hit}}^2}} \quad (6.5)$$

Monte Carlo Based Localization

Mixture(Dual) MCL Algorithm

1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, $Z_m = 5m$, $\Phi = 0.25$
2. Predict
3. **Correct**
4. ~~Preprocess~~
5. **Weigh** (Not efficient in pooling diverse particles in next iteration. Assumed that P1 uses Z_e for W_1 update)
6. Resample

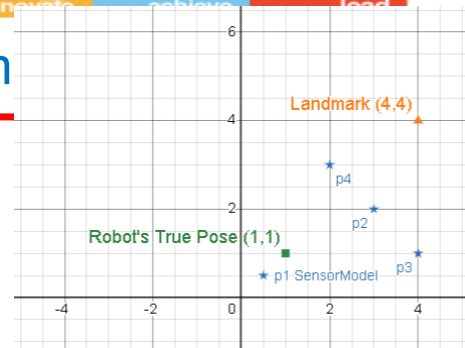
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Table 3
The Mixture-MCL algorithm, here using the third variant of dual MCL (see Table 2)

Algorithm MixtureMCL(X, a, o)

```

 $X' = \emptyset$ 
generate kd-tree  $b$  from  $X$ 
for  $i = 0$  to  $m$  do
  with probability  $1 - \phi$  do
    generate random  $x$  from  $X$  according to  $w_1, \dots, w_m$ 
    generate random  $x' \sim p(x' | a, x)$ 
     $w' = p(o | x')$ 
    add  $(x', w')$  to  $X'$ 
  else do
    generate random  $x' \sim p(x' | o) / \pi(o)$ 
    generate random  $x \sim p(x' | \bar{a}, x) / \pi(x' | a)$ 
     $w' = \pi(o) \pi(x' | a) b(x)$ 
    add  $(x', w')$  to  $X'$ 
  endif
endfor
normalize the importance factors  $w'$  in  $X'$ 
return  $X'$ 
    
```



	P1	P2	P3	P4
Pose X_0	(0.5, 0.5, 0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(0.5, 0.5, 0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	~4.95	2.24	3	2.24
W_1^m	~0.995	0.0005	0.018	0.0005
$\text{Bel}(X_1^m)$	0.9813	0.0005	0.0178	0.0005

(modified for simplicity)

$$\mathcal{N}(z_t^k; z_t^{k*}, \sigma_{\text{hit}}^2) = \frac{1}{\sqrt{2\pi\sigma_{\text{hit}}^2}} e^{-\frac{1}{2} \frac{(z_t^k - z_t^{k*})^2}{\sigma_{\text{hit}}^2}}$$

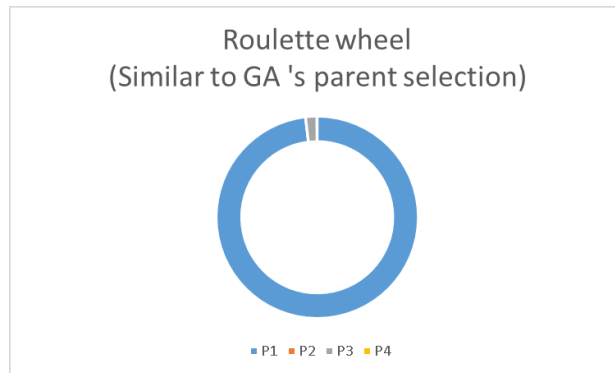
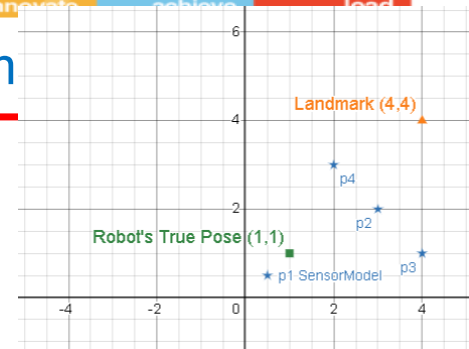
$$h_\theta(x) = \frac{1}{1 + e^{(-\theta^T x)}} \quad (6.5)$$

Monte Carlo Based Localization

Mixture(Dual) MCL Algorithm

1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, Z_m = 5m, Φ = 0.25
2. Predict
3. Correct
4. Preprocess
5. Weigh
6. Resample

One Naïve Estimation Model :
Average of Particles =
(1.4,0.6,0)



	P1	P2	P3	P4
Pose X_0	(0.5, 0.5, 0)	(2,2,0)	(3,1,0)	(1,3,0)
$Bel(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(0.5, 0.5, 0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	~4.95	2.24	3	2.24
W_1^m	~0.995	0.0005	0.018	0.0005
$Bel(X_1^m)$	0.9813	0.0005	0.0178	0.0005

Observe : Though the mixture model is able to handle noisy motion model from influencing the M, due to high W_i of Sensor model based particle P1, its not efficient in pooling diverse particles in next iteration if roulette wheel based mechanism is used.

Solutions: Use AIML to determine the balance between exploration vs exploitation OR use traditional Augmented MCL technique OR use Mixture model ratio to pool for next iteration!!



	P1	P2	P3	P4
X_1	(0.5, 0.5, 0)	(4,1,0)	(0.5, 0.5, 0)	(0.5, 0.5, 0)
$Bel(X_1^m)$	0.25	0.25	0.25	0.25

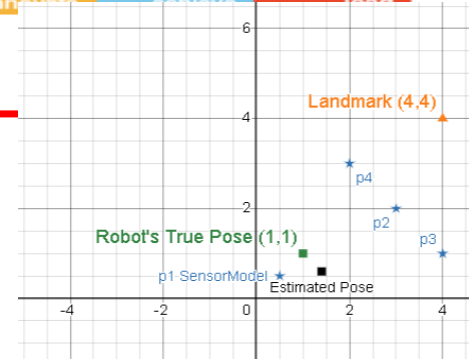
Monte Carlo Based Localization

Idea 2: (Augmented)Mixture(Dual) MCL Algorithm

1. Initialize : M=4, 2D, 5X5, L(4.4), Move 1m east, Sensor noise = 0.5, $Z_m = 5m$, $\Phi = 0.25$
2. Predict
3. Correct
4. Preprocess
5. Weigh
6. Resample

One Naïve Estimation Model :
Average of Particles =
(1.4,0.6,0)

$$W_{avg}^M = \frac{\sum_1^M W(X_1^i)}{4} = \sim 0.2535$$



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Table 3
The Mixture-MCL algorithm, here using the third variant of dual MCL (see Table 2)

Algorithm MixtureMCL(X, a, o)

```

 $X' = \emptyset$ 
generate kd-tree  $b$  from  $X$ 
for  $i = 0$  to  $m$  do
  with probability  $1 - \phi$  do
    generate random  $x$  from  $X$  according to  $w_1, \dots, w_m$ 
    generate random  $x' \sim p(x' | a, x)$ 
     $w' = p(o | x')$ 
    add  $(x', w')$  to  $X'$ 
  else do
    generate random  $x' \sim p(x' | o) / \pi(o)$ 
    generate random  $x \sim p(x' | \bar{a}, x) / \pi(x' | a)$ 
     $w' = \pi(o) \pi(x' | a) b(x)$ 
    add  $(x', w')$  to  $X'$ 
  endif
endif
normalize the importance factors  $w'$  in  $X'$ 
return  $X'$ 
    
```

$$W_{fast} = (1 - \alpha_{fast}) W_{fast} + \alpha_{fast} W_{avg}$$

$$W_{fast} = (1 - 0.5) (0.5) + (0.5) (0.2535) = \sim 0.38$$

$$W_{slow} = (1 - \alpha_{slow}) W_{slow} + \alpha_{slow} W_{avg}$$

$$W_{slow} = (1 - 0.1) (0.9) + (0.1) (0.2535) = \sim 0.84$$

Resampling Effect : Approx. 40% of Particles are Informative.
Retain them ! Need randomization for other 60% of M particles!

	P1	P2	P3	P4
Pose X_0	(0.5, 0.5, 0)	(2,2,0)	(3,1,0)	(1,3,0)
Bel(X_0^m)	0.25	0.25	0.25	0.25
X_1	(0.5, 0.5, 0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	~4.95	2.24	3	2.24
W_1^m	0.995	0.0005	0.018	0.0005
Bel(X_1^m)	0.9813	0.0005	0.0178	0.0005

$$\text{Likelihood Ratio} = \frac{0.38}{0.84} = \sim 0.45$$

$$\text{MAX}\{0, (1 - 0.45)\} = \sim 0.55$$

Hence for M=4 number of particle selection in next round with 0.6 probability (approx. to 60% only for this slide's example), exploration proceeds and for rest exploitation is preferred

Monte Carlo Based Localization

Idea 2: (Augmented)Mixture(Dual) MCL Algorithm

1. Initialize : $M=4$, 2D, 5×5 , $L(4.4)$, Move 1m east, Sensor noise = 0.5, $Z_m = 5m$, $\Phi = 0.25$
2. Predict
3. Correct
4. Preprocess
5. Weigh
6. Resample

One Naïve Estimation Model :
Average of Particles =
(2.75, 1.5, 0)

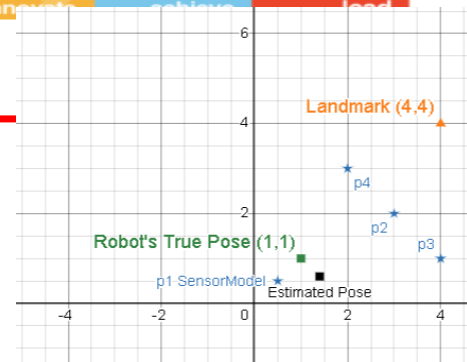


Table 3
The Mixture-MCL algorithm, here using the third variant of dual MCL (see Table 2)

```

Algorithm MixtureMCL( $X, a, o$ )
 $X' = \emptyset$ 
generate kd-tree  $b$  from  $X$ 
for  $i = 0$  to  $m$  do
    with probability  $1 - \phi$  do
        generate random  $x$  from  $X$  according to  $w_1, \dots, w_m$ 
        generate random  $x' \sim p(x' | a, x)$ 
         $w' = p(o | x')$ 
        add  $(x', w')$  to  $X'$ 
    else do
        generate random  $x' \sim p(x' | o) / \pi(o)$ 
        generate random  $x \sim p(x' | \bar{a}, x) / \pi(x' | a)$ 
         $w' = \pi(o) \pi(x' | a) b(x)$ 
        add  $(x', w')$  to  $X'$ 
    endif
endfor
normalize the importance factors  $w'$  in  $X'$ 
return  $X'$ 
    
```

	P1	P2	P3	P4
Pose X_0	(0.5, 0.5, 0)	(2,2,0)	(3,1,0)	(1,3,0)
$\text{Bel}(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(0.5, 0.5, 0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	~4.95	2.24	3	2.24
W_1^m	0.995	0.0005	0.018	0.0005
$\text{Bel}(X_1^m)$	0.9813	0.0005	0.0178	0.0005

Resampling Effect : Approx. 40% of Particles are Informative. Retain them . Resample 2 particles using roulette tech from existing pool (red colored) Need randomization for other 60% of M particles! Ie. Approx 2 particles(green colored)

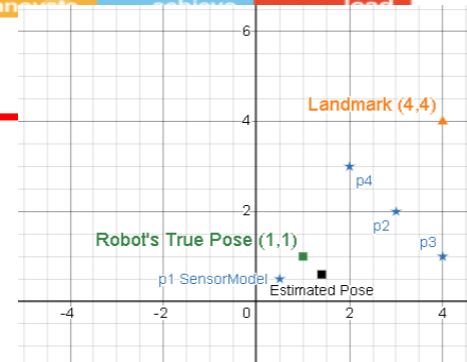
	P1	P2	P3	P4
X_1	(0.5, 0.5, 0)	(4,5,0)	(0.5, 0.5, 0)	(6,0,0)
$\text{Bel}(X_1^m)$	0.25	0.25	0.25	0.25

Monte Carlo Based Localization

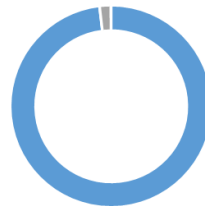
Idea 3: Mixture(Dual) MCL Algorithm

1. Initialize : $M=4$, 2D, 5×5 , $L(4.4)$, Move 1m east, Sensor noise = 0.5, $Z_m = 5m$, $\Phi = 0.25$
2. Predict
3. Correct
4. Preprocess
5. Weigh
6. Resample

One Naïve Estimation Model :
Average of Particles =
(2.63, 1.38, 0)



Roulette wheel
(Similar to GA 's parent selection)



■ P1 ■ P2 ■ P3 ■ P4

Resampling Effect Sample 25% of next iteration particles from sensor model pool. Resample 1 particle using roulette wheel (if many particles like P1 are available) ie., P1 (red colored)
Sample 75% of next iteration particles from motion model Resample 3 particle using roulette wheel ie., P2,3,4 (green colored) . Here old P3 is selected twice and old P4 is selected once.

	P1	P2	P3	P4
Pose X_0	(0.5, 0.5, 0)	(2,2,0)	(3,1,0)	(1,3,0)
$Bel(X_0^m)$	0.25	0.25	0.25	0.25
X_1	(0.5, 0.5, 0)	(3,2,0)	(4,1,0)	(2,3,0)
Z_e	~4.95	2.24	3	2.24
W_1^m	0.995	0.0005	0.018	0.0005
$Bel(X_1^m)$	0.9813	0.0005	0.0178	0.0005

	P1	P2	P3	P4
X_1	(0.5, 0.5, 0)	(4,1,0)	(4,1,0)	(2,3,0)
$Bel(X_1^m)$	0.25	0.25	0.25	0.25

Table 3
The Mixture-MCL algorithm, here using the third variant of dual MCL (see Table 2)

```

Algorithm MixtureMCL( $X, a, o$ )
 $X' = \emptyset$ 
generate kd-tree  $b$  from  $X$ 
for  $i = 0$  to  $m$  do
  with probability  $1 - \phi$  do
    generate random  $x$  from  $X$  according to  $w_1, \dots, w_m$ 
    generate random  $x' \sim p(x' | a, x)$ 
     $w' = p(o | x')$ 
    add  $(x', w')$  to  $X'$ 
  else do
    generate random  $x' \sim p(x' | o) / \pi(o)$ 
    generate random  $x \sim p(x' | \bar{a}, x) / \pi(x' | a)$ 
     $w' = \pi(o) \pi(x' | a) b(x)$ 
    add  $(x', w')$  to  $X'$ 
  endif
endfor
normalize the importance factors  $w'$  in  $X'$ 
return  $X'$ 
    
```

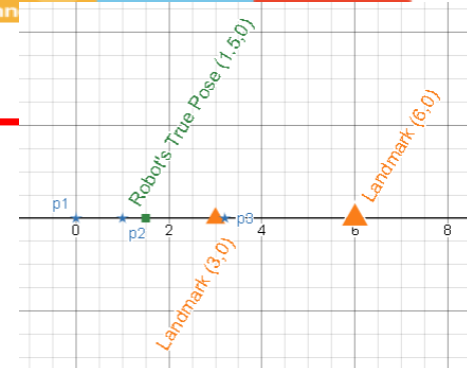
Particle Filter Approach

1. Initialize
 2. Predict
 3. Correct
 4. Preprocess
 5. Weigh
 6. Resample
 7. Repeat
1. Initialize M Particles P_i with uniform belief
 2. Move each P_i based on robot motion model
 3. Compare predicted sensor reading Z_{expected} with actual sensor reading Z_{measured}
 4. Optionally filter out surprising measurements
 5. Weigh each P_i based on importance weights
 6. Resample for particle set for next iteration based on Darwinian approach augmented with short term vs long term likelihood ratio
 7. Reset the belief of particles with equal likelihood

Monte Carlo Based Localization

MCL Algorithm

1. Initialize : $M=3$, 1D, 5X5, L(3 & 6), Move 1m east, $\kappa = 0.5$, $Z_m = 0.5m$
2. Predict : Motion Update
3. Correct : Measurement Update
4. Preprocess
5. Weigh
6. Resample
7. Repeat



	P1	P2	P3
Pose X_1	-1	0	2
$\text{Bel}(X_0^m)$	1/3	1/3	1/3

```

1: Algorithm MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
2:    $\hat{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$ 
3:   for  $m = 1$  to  $M$  do
4:      $x_t^{[m]} = \text{sample\_motion\_model}(u_t, x_{t-1}^{[m]})$ 
5:      $w_t^{[m]} = \text{measurement\_model}(z_t, x_t^{[m]}, m)$ 
6:      $\hat{\mathcal{X}}_t = \hat{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
7:   endfor
8:   for  $m = 1$  to  $M$  do
9:     draw  $i$  with probability  $\propto w_t^{[i]}$ 
10:    add  $x_t^{[i]}$  to  $\mathcal{X}_t$ 
11:   endfor
12:   return  $\mathcal{X}_t$ 

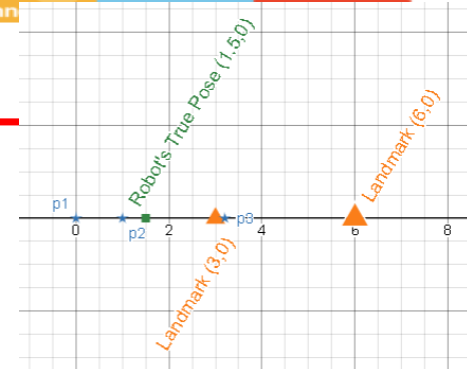
```

Table 8.2 MCL, or Monte Carlo Localization, a localization algorithm based on particle filters.

Monte Carlo Based Localization

MCL Algorithm

1. Initialize : $M=3$, 1D, 5×5 , $L(3 \text{ \& } 6)$, Move 1m east, $\kappa = 0.5$, $Z_m = 0.5m$
2. **Predict : Motion Update**
3. **Correct : Measurement Update**
4. Preprocess
5. Weigh
6. Resample
7. Repeat



	P1	P2	P3
Pose X_1	0	1	3
$\text{Bel}(X_0^m)$	1/3	1/3	1/3

```

1: Algorithm MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
2:    $\mathcal{X}_t = \mathcal{X}_{t-1} = \emptyset$ 
3:   for  $m = 1$  to  $M$  do
4:      $x_t^{[m]} = \text{sample\_motion\_model}(u_t, x_{t-1}^{[m]})$ 
5:      $w_t^{[m]} = \text{measurement\_model}(z_t, x_t^{[m]}, m)$ 
6:      $\mathcal{X}_t = \mathcal{X}_t + \{x_t^{[m]}, w_t^{[m]}\}$ 
7:   endfor
8:   for  $m = 1$  to  $M$  do
9:     draw  $i$  with probability  $\propto w_t^{[i]}$ 
10:    add  $x_t^{[i]}$  to  $\mathcal{X}_t$ 
11:   endfor
12:   return  $\mathcal{X}_t$ 
    
```

Table 8.2 MCL, or Monte Carlo Localization, a localization algorithm based on particle filters.

(modified for simplicity)

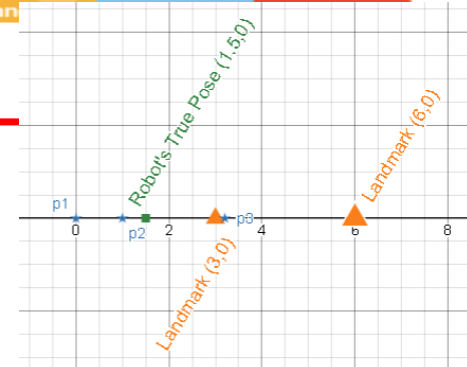
$$\begin{pmatrix} x_t \\ y_t \\ \theta_t \end{pmatrix} = \begin{pmatrix} x_{t-1} \\ y_{t-1} \\ \theta_{t-1} \end{pmatrix} + \begin{pmatrix} v \cos \theta \Delta t \\ v \sin \theta \Delta t \\ w \Delta t \end{pmatrix}$$

Linear Regression Model $x_t = x_{t-1} + 1$ 35

Monte Carlo Based Localization

MCL Algorithm

1. Initialize : $M=3$, 1D, 5×5 , $L(3 \text{ \& } 6)$, Move 1m east, $\lambda = 0.5$, $Z_m = 0.5m$
2. Predict : Motion Update
3. **Correct : Measurement Update**
4. Preprocess
5. Weigh
6. Resample
7. Repeat



	P1	P2	P3
Pose X_1	0	1	3
$\text{Bel}(X_0^m)$	1/3	1/3	1/3
Z_e	3	2	3

```

1: Algorithm MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
2:    $\tilde{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$ 
3:   for  $m = 1$  to  $M$  do
4:      $x_t^{[m]} = \text{sample\_motion\_model}(u_t, x_{t-1}^{[m]})$ 
5:      $w_t^{[m]} = \text{measurement\_model}(z_t, x_t^{[m]}, m)$ 
6:      $\tilde{\mathcal{X}}_t = \tilde{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
7:   endfor
8:   for  $m = 1$  to  $M$  do
9:     draw  $i$  with probability  $\propto w_t^{[i]}$ 
10:    add  $x_t^{[i]}$  to  $\mathcal{X}_t$ 
11:  endfor
12:  return  $\mathcal{X}_t$ 

```

$$p(z_t^k | x_t, m) = \begin{pmatrix} z_{\text{hit}} \\ z_{\text{short}} \\ z_{\text{max}} \\ z_{\text{rand}} \end{pmatrix}^T \cdot \begin{pmatrix} p_{\text{hit}}(z_t^k | x_t, m) \\ p_{\text{short}}(z_t^k | x_t, m) \\ p_{\text{max}}(z_t^k | x_t, m) \\ p_{\text{rand}}(z_t^k | x_t, m) \end{pmatrix}$$

- Used Manhattan Distance
- Restricted view measured w.r.t 2nd landmark from P3³⁶

Table 8.2 MCL, or Monte Carlo Localization, a localization algorithm based on particle filters.

Monte Carlo Based Localization

1. Initialize : $M=3$, 1D, 5×5 , $L(3 \text{ \& } 6)$, Move 1m east, $\kappa = 0.5$, $Z_m = 0.5m$
2. Predict : Motion Update
3. Correct : Measurement Update
4. **Preprocess**
5. **Weigh**
6. Resample
7. Repeat

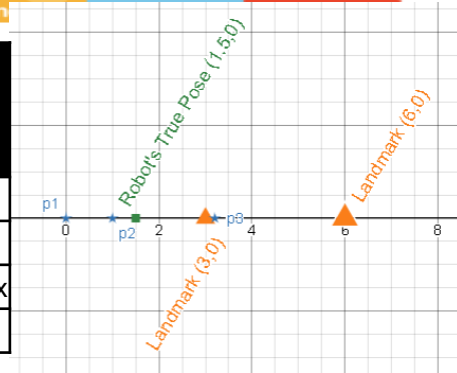
```

1: Algorithm MCL( $\mathcal{X}_{t-1}, u_t, z_t, m$ ):
2:    $\hat{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$ 
3:   for  $m = 1$  to  $M$  do
4:      $x_t^{[m]} = \text{sample\_motion\_model}(u_t, x_{t-1}^{[m]})$ 
5:      $w_t^{[m]} = \text{measurement\_model}(z_t, x_t^{[m]}, m)$ 
6:      $\hat{\mathcal{X}}_t = \hat{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
7:   endfor
8:   for  $m = 1$  to  $M$  do
9:     draw  $i$  with probability  $\propto w_t^{[i]}$ 
10:    add  $x_t^{[i]}$  to  $\hat{\mathcal{X}}_t$ 
11:   endfor
12:   return  $\hat{\mathcal{X}}_t$ 

```

Table 8.2 MCL, or Monte Carlo Localization, a localization algorithm based on particle filters.

	Mixture weights s $P(C)$	Measurement Likelihood $P(Z=0.5 C)$	
Hit	0.8	0.01	$Z_m = Z_e$
Short	0.1	0.95	$Z_m < Z_e$
Max	0.05	0.01	$Z_m = \text{Max}$
Rand	0.05	0.2	Others

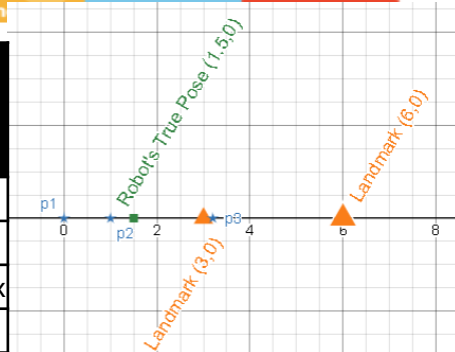


	P1	P2	P3
Pose X_1	0	1	3
$\text{Bel}(X_0^m)$	1/3	1/3	1/3
Z_e	3	2	3
May be	Short	Short	Short

Monte Carlo Based Localization

1. Initialize : $M=3$, 1D, 5×5 , $L(3 \text{ \& } 6)$, Move 1m east, $\kappa = 0.5$, $Z_m = 0.5m$
2. Predict : Motion Update
3. Correct : Measurement Update
4. **Preprocess**
5. **Weigh**
6. Resample
7. Repeat

	Mixture weight s $P(C)$	Measurement Likelihood $P(Z=0.5 C)$	
Hit	0.8	0.01	$Z_m = Z_e$
Short	0.1	0.95	$Z_m < Z_e$
Max	0.05	0.01	$Z_m = \text{Max}$
Rand	0.05	0.2	Others



	P1	P2	P3
Pose X_1	0	1	3
$\text{Bel}(X_0^m)$	1/3	1/3	1/3
Z_e	3	2	3
May be	Short	Short	Short
$P(C_{\text{short}} Z_m)$	$0.1 \cdot 0.95 \cdot 0.33$	~ 0.0314	~ 0.0314

```

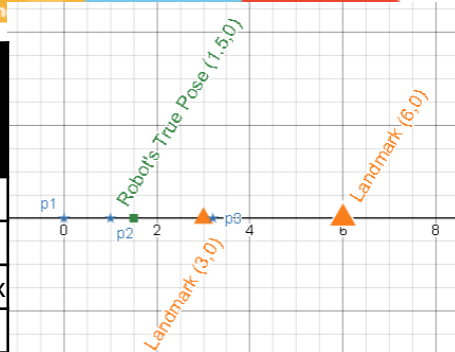
1: Algorithm test_range_measurement( $z_t^k, \hat{x}_t, m$ ):
2:    $p = q = 0$ 
3:   for  $m = 1$  to  $M$  do
4:      $p = p + z_{\text{short}} \cdot p_{\text{short}}(z_t^k | x_t^{[m]}, m)$ 
5:      $q = q + z_{\text{hit}} \cdot p_{\text{hit}}(z_t^k | x_t^{[m]}, m) + z_{\text{short}} \cdot p_{\text{short}}(z_t^k | x_t^{[m]}, m)$ 
6:      $+ z_{\text{max}} \cdot p_{\text{max}}(z_t^k | x_t^{[m]}, m) + z_{\text{rand}} \cdot p_{\text{rand}}(z_t^k | x_t^{[m]}, m)$ 
7:   endfor
8:   if  $p/q \leq \chi$  then
9:     return accept
10:  else
11:    return reject
12:  endif
    
```

$$p(\bar{c}_t^k = \text{short} | z_t^k, z_{1:t-1}, u_{1:t}, m) = \frac{\int p_{\text{short}}(z_t^k | x_t, m) z_{\text{short}} \bar{\text{bel}}(x_t) dx_t}{\int \sum_c p_c(z_t^k | x_t, m) z_c \bar{\text{bel}}(x_t) dx_t} \quad (8.10)$$

Monte Carlo Based Localization

1. Initialize : $M=3$, 1D, 5X5, L(3 & 6), Move 1m east, $\lambda = 0.5$, $Z_m = 0.5m$
2. Predict : Motion Update
3. Correct : Measurement Update
4. **Preprocess**
5. **Weigh**
6. Resample
7. Repeat

	Mixture weight s P(C)	Measurement Likelihood P(Z=0.5 C)	
Hit	0.8	0.01	$Z_m = Z_e$
Short	0.1	0.95	$Z_m < Z_e$
Max	0.05	0.01	$Z_m = \text{Max}$
Rand	0.05	0.2	Others



$p = \sim 0.09$
 $q = \text{Denominator} = \sim 0.11$
 $p/q = \sim 0.84$
 $p/q > \lambda$

Inference :

- Reject the Z_m . Z. Stop this iteration. No resampling.
- Use the same particles again to begin next iteration.

	P1	P2	P3
Pose X_1	0	1	3
$\text{Bel}(X_0^m)$	1/3	1/3	1/3
Z_e	3	2	3
May be	Short	Short	Short
$P(C_{\text{short}} Z_m)$	$0.1 \cdot 0.95 \cdot 0.33$	~ 0.0314	~ 0.0314

$$p(\bar{c}_t^k = \text{short} | z_t^k, z_{1:t-1}, u_{1:t}, m) = \frac{\int p_{\text{short}}(z_t^k | x_t, m) z_{\text{short}} \bar{\text{bel}}(x_t) dx_t}{\int \sum_c p_c(z_t^k | x_t, m) z_c \bar{\text{bel}}(x_t) dx_t} \quad (8.10)$$

```

1: Algorithm test_range_measurement( $z_t^k, \mathcal{X}_t, m$ ):
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7:   endfor
8:   if  $p/q \leq \lambda$  then
9:     return accept
10:  else
11:    return reject
12:  endif
    
```

Particle Filter Approach

1. Initialize

2. Predict

3. Correct

4. Preprocess

5. Weigh

6. Resample

7. Repeat

1. Initialize M Particles P_i with uniform belief

2. Move each P_i based on robot motion model

3. Compare predicted sensor reading Z_{expected} with actual sensor reading Z_{measured}

4. Optionally filter out surprising measurements

5. Weigh each P_i based on importance weights

6. Resample for particle set for next iteration based on Darwinian approach augmented with short term vs long term likelihood ratio

7. Reset the belief of particles with equal likelihood

CAN BE APPLIED TO GRID BASED LOCALIZATION AS WELL WITH CAREFUL CUSTOMIZATION

Required Reading

- T1 Chapter 4.2: 4.2.1, 4.2.2, 4.2.4, 8.3: 8.3.1, 8.3.2, 8.3.3, 8.3.4, 8.4

Next Class Plan :

- ✓ Mapping
- ✓ Occupancy Grid Mapping